Automated Segmentation of Corneal Structures and Keratitis Lesions in AS-OCT Images

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Abstract

Anterior Segment Optical Coherence Tomography (AS-OCT) provides high-resolution, non-invasive cross-sectional imaging of the cornea and is well suited for quantitative assessment of infectious keratitis. However, automated lesion segmentation in keratitis AS-OCT remains underexplored, and no published reproducible deep learning benchmark had previously been established for lesion segmentation on the publicly available AS-OCT Image Dataset for Keratitis (AIDK). This project presents a controlled three-architecture benchmark for segmenting cornea, keratitis lesion, and iris from AIDK images using an adapted ScLNet model, a ResNet-34 U-Net baseline, and a TransUNet hybrid model. All three architectures were trained and evaluated under matched data splits, preprocessing, augmentation, optimization, and evaluation settings so that only architectural differences varied across experiments. The study uses the public AIDK dataset, which contains 1,168 AS-OCT B-scans from 64 patients, including a full-frame subset of 768 images with cornea, lesion, and iris annotations used for the final benchmark. Across the three models, lesion segmentation converged to a narrow performance range around a Dice score of 0.67, with a spread of only 0.012, while cornea and iris segmentation achieved consistently strong performance. These results are consistent with a data-limited performance ceiling on the current public dataset, although the shared use of a ResNet-34 encoder, training from scratch, and single-seed runs per model prevent a fully clean separation between data-driven and architectural effects. The project presents what appears to be the first publicly reproducible benchmark for keratitis lesion segmentation on AIDK and suggests that future progress on this dataset is likely to depend on improved data quality, annotation consistency, and additional labeled samples, alongside further exploration of pretrained encoders and alternative training regimes.

1 Background

Infectious keratitis is a severe ocular infection that can rapidly lead to corneal scarring and vision loss. Global estimates indicate that corneal opacity is the fifth leading cause of blindness and is responsible for approximately 1.5–2 million new cases of monocular blindness each year [1]. The incidence of infectious keratitis varies widely (2.5–799 cases per 100 000 population) and is influenced by risk factors such as contact lens wear, ocular trauma, ocular surface disease, lid disease and postoperative complications [1]. Taken together, these statistics show why timely diagnosis and careful quantitative assessment of keratitis lesions matter in both clinical and socioeconomic terms.

Optical coherence tomography (OCT) has revolutionized ophthalmology by providing non-invasive, high-resolution cross-sectional imaging of ocular tissues. OCT employs low-coherence visible or near-infrared light to produce micrometer-scale images of the cornea, retina and other structures [2]. Advances from time-domain to spectral-domain and swept-source systems have improved acquisition speed and penetration, enabling volumetric imaging of the anterior segment (AS-OCT) and facilitating detailed evaluation of corneal and scleral morphology [3]. Unlike slit-lamp biomicroscopy, AS-OCT can visualize stromal depth and lesion extent, which is essential for monitoring infectious keratitis and guiding treatment. However, manual delineation of structures in OCT images is laborious and subjective; automated segmentation is necessary for quantitative analysis and reproducible clinical research.

Deep learning has become a popular solution for medical image segmentation. Convolutional networks such as U-Net and its variants have demonstrated state-of-the-art performance across a range of biomedical applications. A recent survey highlights key trends driving progress, including multi-scale analysis, attention mechanisms, integration of prior knowledge, semi- and unsupervised learning, multimodality fusion, probabilistic segmentation and foundation models [4]. In ophthalmology, specialized architectures have been developed for corneal OCT segmentation. CorneaNet segments three corneal layers and achieves over 99% accuracy on healthy and keratoconic eyes [5], while ScLNet extends a ResNet backbone with multi-scale inputs, dense atrous convolutions and a boundary-aware decoder to segment scleral lens, tear reservoir and cornea [6]. These successes demonstrate that deep networks can precisely delineate corneal structures under controlled conditions, yet there is little evidence on performance for infectious lesions. Published studies report moderate lesion Dice scores (≈ 0.79) even with additional training data [7], and there has been no publicly available benchmark for keratitis segmentation.

To address this gap, this project uses the publicly available AS-OCT Image Dataset for Keratitis (AIDK) [7] to establish a controlled benchmark for keratitis lesion segmentation. The dataset provides pixel-level annotations for cornea, lesion, and, in the full-frame subset, iris across 1,168 images from 64 patients [7]. Three segmentation architectures were evaluated under matched experimental conditions: (i) an adaptation of ScLNet, (ii) a baseline ResNet-34 U-Net, and (iii) a hybrid TransUNet combining CNN encoders with transformer bottlenecks.

By holding the training pipeline constant and varying only the model architecture, this study examines whether the current lesion-segmentation performance limit on AIDK is primarily architectural or data-driven. The results show that all three models converge to a similar lesion Dice score of approximately 0.67 under these shared training conditions, a pattern consistent with a data-driven performance ceiling, although confounds such as the absence of pretraining and the shared encoder backbone prevent a clean separation of data-driven from architectural effects.

2 Dataset

2.1 Acquisition and Cohort

The AIDK dataset comprises 1 168 anterior-segment OCT B-scans acquired from 64 keratitis patients using a Tomey Casia SS-1000 swept-source AS-OCT system [7]. Patients ranged in age from 4 to 92 years, and scans were collected between September 2018 and March 2023 [7]. Each B-scan depicts a cross-section of the cornea and anterior segment; two acquisition modes are represented. The *partial-frame* subset (58 patients, 400 images) uses a zoomed field of view focused on the central cornea, with images at either 1657×1000 or 1723×1000 pixel resolution depending on the acquisition machine, while the *full-frame* subset (6 patients, 768 images) captures the entire anterior segment at 1723×1000 pixel resolution [7]. Both subsets were acquired with the same device, which uses swept-source technology to achieve high axial resolution and deep penetration [7]. The distribution of images across subsets and resolutions is summarized in Table 1.

Table 1: Summary of the AIDK dataset. Each B-scan includes cornea and lesion masks; iris masks are available only for the full-frame subset. Partial-frame images appear at two resolutions depending on the acquisition machine.

Subset	**Patients**	**Images**	**Resolution (pixels)**
Partial-frame	58	400	1657 × 1000 or 1723 × 1000
Full-frame	6	768	1723 × 1000
Total	64	1,168	–

2.2 Annotation Protocol

Each B-scan was manually annotated by three experienced ophthalmic clinicians using Labelme, delineating the cornea and keratitis lesion in both subsets and the iris in the full-frame subset [7]. For quality control, consensus masks were generated from the overlapping regions of the annotators. Inter-annotator agreement was measured using the Dice similarity coefficient and intraclass correlation coefficient (ICC). The mean Dice coefficients were > 0.96 for the cornea, > 0.90 for the iris and > 0.83 for the lesion, and the corresponding ICCs were 0.897, 0.900 and 0.708 respectively [7]. These high agreement values indicate reliable delineation of corneal and iris boundaries but reflect moderate variability in lesion borders, highlighting the intrinsic difficulty of keratitis segmentation.

2.3 Data Preparation for Experiments

For the experiments in this study, only the full-frame subset was used for model training and evaluation because it includes iris annotations, providing complete anatomical context. The partial-frame subset was held out due to the absence of iris labels; combining the subsets without removing the iris class introduces a label conflict that significantly degrades iris segmentation performance (see Section 4). Images and masks were resized to 512×512 pixels and normalized. A patient-level random split divided the full-frame images into training (70%), validation (20%), and test (10%) sets with a fixed random seed to ensure reproducibility. Four classes were defined: background, cornea, lesion and iris. Table 2 summarizes the number of images in each split.

Table 2: Number of full-frame images in each subset after resizing and patient-level splitting.

Subset	**Images**	**Percentage**
Training	537	70%
Validation	154	20%
Test	77	10%
Total	768	100%

Figure 1 shows a full-frame B-scan with the expert - annotation overlay segmentation from the AIDK dataset. The cornea (yellow), keratitis lesion (red) and iris (blue) are highlighted for illustration.

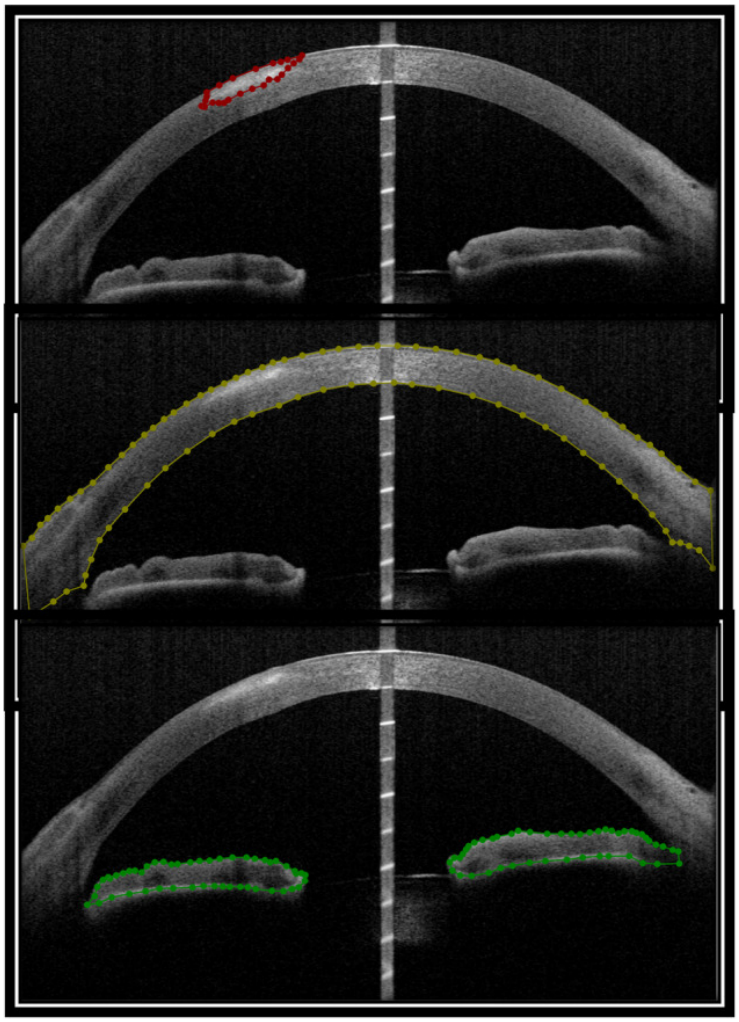

Figure 1: AS-OCT B-scan showing the cornea (yellow), keratitis lesion (red) and iris (blue) in the AIDK dataset.

3 Methods

Three segmentation architectures are evaluated. All models were implemented in PyTorch and trained with the same experimental setup to ensure a controlled comparison of architectural effects.

3.1 Shared Training Pipeline

To ensure that performance differences can be attributed solely to architectural choices, all three models share an identical training pipeline. The AIDK full-frame subset (768 images from 6 patients) was used for all experiments, with iris segmentation included as a third foreground class (background, cornea, lesion, iris). Data was split into training (70%), validation (20%), and test (10%) sets using a fixed random seed (SEED = 42) to guarantee identical splits across all experiments.

All images and corresponding annotation masks were resized to 512×512 pixels. Training augmentations followed the ScLNet paper's protocol [6] (random vertical flips, translations ±10%, rotations ±10°, and scaling 0.9–1.1), extended with Gaussian noise and random brightness/contrast adjustments to improve robustness. The full augmentation set was: random vertical flips ($p = 0.5$), affine transformations (translation ±10%, scale 0.9–1.1, rotation ±10°, $p = 0.7$), Gaussian noise ($p = 0.3$), and random brightness/contrast adjustments ($p = 0.3$), implemented using the Albumentations library [8]. No augmentation was applied during validation or testing.

All models were trained with stochastic gradient descent (SGD) with momentum 0.9 and weight decay 10^{-4}, using an initial learning rate of 0.01 reduced by a factor of 10 every 20 epochs (step-decay schedule). Training ran for up to 100 epochs with early stopping triggered after 50 consecutive epochs without validation loss improvement. Mixed-precision training (AMP) [9] was enabled for all models to reduce GPU memory requirements without measurable impact on convergence. Training was conducted on Kaggle's NVIDIA Tesla T4 GPU instances.

3.2 Model 1: ScLNet (Adapted)

The first model is an adaptation of ScLNet [6], originally designed for segmenting scleral lens, tear fluid reservoir, and cornea from OCT images. ScLNet was selected because it is the only published architecture specifically designed for corneal OCT layer segmentation, and its boundary-aware multi-task design appeared well-suited to the challenge of segmenting small, ambiguous keratitis lesions.

ScLNet's encoder is based on a ResNet-34 backbone [10] with five stages of stacked residual blocks; in the reimplementation, channel widths 64, 128, 256, 512, 512 with block counts 3, 4, 6, 3, 3 were used. A feature subtraction operation between adjacent encoder layers emphasizes boundary information by computing the element-wise difference between feature maps before and after parallel convolutional processing. The resulting difference features, which capture high-frequency boundary signals, are passed to the boundary decoder as skip connections.

A multi-scale input pyramid feeds the original image at four resolutions (full, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$) into the encoder at corresponding stages. At each stage, the downsampled input is adapted to the stage's channel width via a convolutional block and merged with the encoder features through Dense Atrous Convolution (DAC) blocks [11]. DAC blocks employ cascaded branches with dilation rates of 1, 3, and 5 to capture multi-scale contextual information without increasing parameter count [12], enabling the model to handle structures of varying sizes.

The decoder follows a dual-task architecture. The boundary detection decoder receives boundary-emphasized features and produces a boundary prediction map. The segmentation decoder receives standard encoder features as well as intermediate features from the boundary decoder via cross-decoder skip connections, enabling boundary-aware refinement of region predictions. Both decoders use upsampling followed by concatenation with skip connections and pairs of residual blocks at each level. The final segmentation and boundary outputs are produced by 1×1 convolutional layers followed by a $4\times$ bilinear upsampling to restore the original resolution.

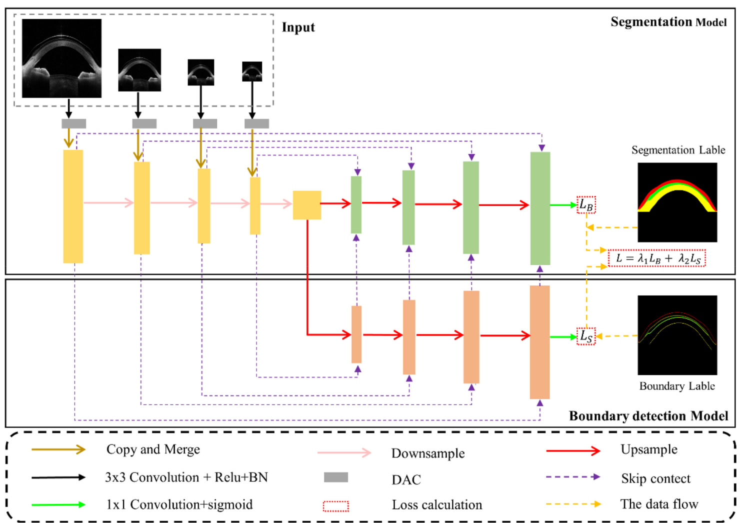

Figure 2: Architecture of the adapted ScLNet model. The encoder employs a ResNet-34 backbone with multi-scale input injection via Dense Atrous Convolution (DAC) blocks and boundary-emphasizing feature subtraction. The dual-task decoder produces both segmentation masks (primary task) and boundary predictions (auxiliary task), with cross-decoder skip connections allowing boundary features to refine the segmentation output. The total loss is a weighted combination of per-class log-Dice losses from both tasks ($\lambda = 0.4$ for boundary, $1 - \lambda = 0.6$ for segmentation).

The combined loss function follows the paper's weighting formulation:

$$\mathcal{L}_{\text{total}} = \lambda \, \mathcal{L}_{\text{boundary}} + (1 - \lambda) \, \mathcal{L}_{\text{segmentation}} \tag{1}$$

where $\lambda = 0.4$. In the reimplementation, a per-class independent log-Dice formulation was adopted for both $\mathcal{L}_{\text{boundary}}$ and $\mathcal{L}_{\text{segmentation}}$. Specifically, for each class c, the Dice coefficient is computed independently as:

$$D_c = \frac{2\sum_i p_{c,i}\, g_{c,i} + \epsilon}{\sum_i p_{c,i}^2 + \sum_i g_{c,i}^2 + \epsilon} \tag{2}$$

where $p_{c,i}$ and $g_{c,i}$ are the predicted softmax probability and ground truth for class c at pixel i, and $\epsilon = 1.0$ is a smoothing constant. The per-class log-Dice loss is then $-\log(D_c)$, and the total segmentation loss is the mean of per-class losses across all classes including background. The total model contains approximately 92.6 million trainable parameters.

3.3 Model 2: ResNet-34 UNet (Baseline)

The second model is a ResNet-34 encoder paired with a standard UNet decoder [13]. This model was selected as a practical baseline because the AIDK dataset authors reported that they trained an "improved U-Net" for keratitis lesion segmentation and obtained a Dice score of 0.79 using the AIDK dataset together with an additional supplementary dataset [7]. However, the architecture and training details of that improved U-Net were not published in sufficient detail to permit direct reproduction. As a result, a simple and well-established UNet-style architecture was used as a transparent baseline for comparison.

The encoder is implemented with a ResNet-34 backbone. This provides a strong and widely used feature extractor while keeping the baseline architecture substantially simpler than the adapted ScLNet model. Unlike ScLNet, this baseline does not include a multi-scale input pyramid, dense atrous convolution blocks, feature subtraction, or a boundary-supervision branch. The goal is not to reproduce the unpublished AIDK model exactly, but rather to establish a clear conventional baseline against which the more specialized architectures can be evaluated.

The decoder consists of four upsampling stages. Each stage performs $2\times$ bilinear upsampling, concatenates the corresponding encoder skip connection, and applies two residual blocks. A final $4\times$ upsampling restores the original spatial resolution, and a 1×1 convolution produces the class logits.

Because this UNet variant predicts only segmentation masks, the loss function is simplified to a single per-class independent log-Dice loss (Equation 2) applied to the segmentation output alone. The model contains approximately 30 million trainable parameters, making it substantially smaller than the adapted ScLNet and useful as a conventional baseline in the three-model comparison.

3.4 Model 3: TransUNet Hybrid

The third model is a novel hybrid architecture designed specifically to investigate whether the lesion segmentation ceiling observed in Models 1 and 2 could be broken by incorporating global contextual reasoning. Its design is motivated by three hypotheses about the failure modes of the CNN-based models:

1. **Limited effective receptive field;** While deep CNNs have large theoretical receptive fields, Luo et al. [14] showed that the *effective* receptive field, the region that actually influences predictions, is much smaller. For lesion segmentation, where identifying a small abnormality may require reasoning about the surrounding tissue context (e.g., corneal curvature, relative position of the iris), this limitation may be significant.

2. **Unsupervised boundary precision.** ScLNet's boundary decoder provided small but consistent improvements in Hausdorff Distance (HD) compared to the UNet baseline, suggesting that boundary-aware features help with spatial precision, even if region overlap metrics (DSC) are unaffected.

3. **Symmetric handling of false positives and false negatives.** Standard Dice loss penalizes false positives and false negatives equally. For clinical lesion segmentation, failing to detect a lesion (false negative) is arguably more harmful than over-segmenting one (false positive).

3.4.1 Architecture

The encoder uses the same ResNet-34 backbone as Models 1 and 2 for fair comparison. Boundary-aware feature subtraction is incorporated into the skip connections: at each encoder stage, the feature maps pass through a parallel 3×3 convolutional layer, and the difference between the original and parallel-processed features is concatenated with the original features before being passed to the decoder. This doubles the channel width of each skip connection and is the one ScLNet component that demonstrated measurable benefit (improved HD) in the experiments.

At the bottleneck (deepest encoder output, spatial resolution 16×16 for 512×512 inputs), the feature maps are flattened into a sequence of 256 tokens, each representing a 32×32 pixel region. A transformer encoder [15,16] processes this sequence through four layers of multi-head self-attention (8 heads) and feed-forward networks (MLP ratio 4), with learned positional embeddings and layer normalization. The transformer output is reshaped back to spatial dimensions and passed to the decoder.

The decoder follows the same UNet-style structure as Model 2 (four up-sampling stages with skip connections and residual blocks) but receives the enriched skip connections (with concatenated boundary subtraction features). Four auxiliary segmentation heads are attached at each decoder level for deep supervision [17], providing direct gradient signal to intermediate features during training. During inference, only the main output head is used.

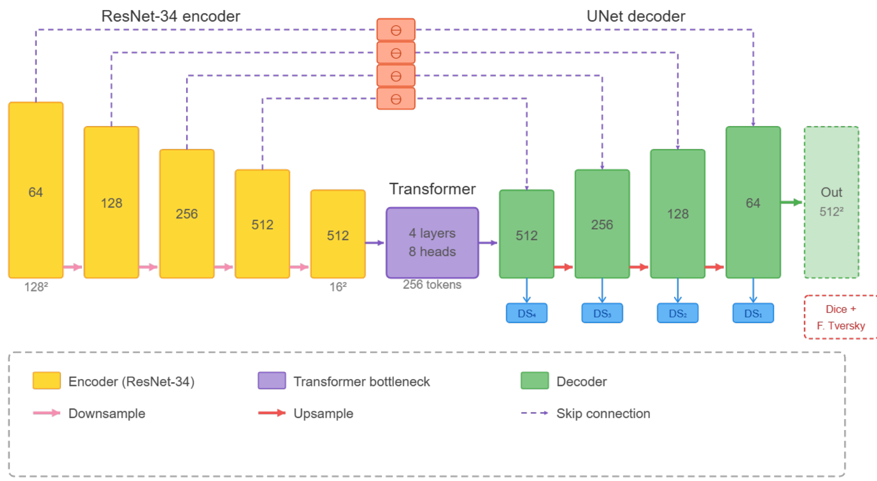

Figure 3: Architecture of the proposed TransUNet Hybrid model. The encoder follows a standard ResNet-34 backbone, with boundary-aware feature subtraction applied at each skip connection (element-wise difference between original and parallel-convolved features, concatenated to the original). A four-layer transformer encoder processes the deepest feature map as a 256-token sequence with learned positional embeddings, enabling global self-attention across the full spatial extent of the image. The UNet-style decoder receives enriched skip connections and produces segmentation predictions at each level for deep supervision during training. The model uses a hybrid loss: per-class log-Dice for cornea, iris, and background, and Focal Tversky loss for the lesion class.

3.4.2 Loss Function

The TransUNet Hybrid uses a class-specific hybrid loss. For the cornea, iris, and background classes, the standard per-class log-Dice loss (Equation 2) is used, as these classes are already well-segmented by the baseline models. For the lesion class, Focal Tversky Loss [18, 19] is used instead:

$$TI_c = \frac{\sum_i p_{c,i}\, g_{c,i} + \epsilon}{\sum_i p_{c,i}\, g_{c,i} + \alpha \sum_i p_{c,i}(1 - g_{c,i}) + \beta \sum_i (1 - p_{c,i}) g_{c,i} + \epsilon} \tag{3}$$

$$\mathcal{L}_{\text{FTL}} = (1 - TI_c)^{\gamma} \tag{4}$$

where $\alpha = 0.3$ controls the false positive penalty, $\beta = 0.7$ controls the false negative penalty (with $\alpha + \beta = 1$), and $\gamma = \frac{4}{3}$ is the focusing parameter following the recommendation of Abraham and Khan [18]. Setting $\beta > \alpha$ biases the model toward higher recall on the lesion class. The total loss for the main output is the mean of per-class losses across all four classes. For deep supervision, the same loss is computed at each decoder level using downsampled ground truth, with weights [0.1, 0.2, 0.3, 0.4] (coarser levels receive less weight) added to the main loss. The total model contains approximately 75 million trainable parameters.

3.5 Evaluation Metrics

Model performance was evaluated on the held-out test set (77 images) using five metrics, following the evaluation protocol of Cao et al. [6]:

- **Dice Similarity Coefficient (DSC)**:

$$\mathrm{DSC} = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

- **Intersection over Union (IoU)**:

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}$$

- **Matthews Correlation Coefficient (MCC)**:

$$\mathrm{MCC} = \frac{\mathrm{TP}\,\mathrm{TN} - \mathrm{FP}\,\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}$$

- **Precision**:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

- **Hausdorff Distance (HD)**:

$$\mathrm{HD}(A, B) = \max\left(\sup_{a \in A} \inf_{b \in B} \|a - b\|,\ \sup_{b \in B} \inf_{a \in A} \|b - a\| \right)$$

All metrics are computed per-class (excluding background) and reported as the mean across all test images. For Hausdorff Distance, cases where one set is empty (complete miss or false alarm on an entire image) are excluded from the mean, and the count of excluded cases is reported separately to capture detection reliability.

4 Results

4.1 ScLNet: Initial Results and Optimization Attempts

The adapted ScLNet achieved strong performance on cornea and iris segmentation, with test-set DSC of 0.9705 and 0.9288 respectively. Lesion segmentation, however, plateaued at a DSC of 0.6713, with an IoU of 0.6068. Table 3 presents the complete per-class test-set results.

Table 3: ScLNet test-set performance on the AIDK keratitis dataset (full-frame subset, four classes including iris).

Class	DSC	IoU	MCC	Precision	HD
Cornea	0.9705	0.9428	0.9675	0.9658	16.35
Lesion	0.6713	0.6068	0.6721	0.6788	7.30
Iris	0.9288	0.8685	0.9269	0.9199	13.58

Several attempts were made to improve lesion segmentation performance within the ScLNet framework:

Class weighting experiments. Explicit class weights of 3.0 and 5.0 were applied to the lesion class in the log-Dice loss. All weighting schemes degraded cornea and iris performance (e.g., at weight 5.0, cornea DSC dropped from 0.97 to 0.85 and iris DSC from 0.93 to 0.72) while lesion DSC remained in the 0.59–0.62 range, below the unweighted result of 0.67. Table 4 summarizes these results. This finding is consistent with the observation that per-class independent log-Dice already provides structural handling of class imbalance, and adding explicit weights amounts to over-correction.

Table 4: Effect of explicit lesion class weighting on ScLNet performance. The unweighted configuration achieved the best lesion DSC (0.67), while increased weighting degraded cornea and iris performance without improving lesion segmentation.

Lesion Weight	Cornea DSC	Lesion DSC	Iris DSC
1.0 (unweighted)	0.9705	0.6713	0.9288
3.0	0.9400	0.6100	0.8900
5.0	0.8500	0.6200	0.7200

Dataset configuration experiments. To maximize lesion patient diversity (58 patients in the partial-frame subset versus 6 in the full-frame subset), both subsets were combined for training. However, this revealed an iris label conflict: partial-frame images do not contain iris annotations, causing visible iris tissue to be labeled as background. When both subsets were combined with iris included as a segmentation class, iris DSC collapsed from 0.93 to 0.48, confirming the label conflict hypothesis. The recommended configuration for future work using both subsets is `INCLUDE_IRIS=False`, which avoids the conflict at the cost of excluding iris as a segmentation target.

4.2 Three-Architecture Comparison

Table 5 presents the primary result of this study: a controlled comparison of all three architectures on the same test set under identical training conditions.

Table 5: Complete per-class evaluation metrics for all three architectures on the AIDK keratitis test set. All models demonstrate strong performance on cornea and iris segmentation (DSC > 0.92) while converging to similar lesion DSC in the 0.66–0.67 range. The UNet baseline achieves the highest lesion precision (0.6791), while ScLNet achieves the best lesion HD (7.30 pixels).

Model	**Class**	**DSC**	**IoU**	**MCC**	**Precision**	**HD**
ScLNet	Cornea	0.9705	0.9428	0.9675	0.9658	16.35
	Lesion	0.6713	0.6068	0.6721	0.6788	7.30
	Iris	0.9288	0.8685	0.9269	0.9199	13.58
ResNet-34 UNet	Cornea	0.9696	0.9411	0.9665	0.9653	21.01
	Lesion	0.6686	0.6029	0.6698	0.6791	7.96*
	Iris	0.9276	0.8662	0.9257	0.9226	13.89
TransUNet Hybrid	Cornea	0.9687	0.9395	0.9656	0.9686	17.48
	Lesion	0.6585	0.5870	0.6610	0.6241	8.19*
	Iris	0.9269	0.8651	0.9250	0.9187	13.63

*HD computed on 76/77 images; 1 image excluded due to empty prediction.

On region overlap metrics (DSC, IoU, MCC), the three models are functionally equivalent. The maximum difference in lesion DSC is 0.012 (ScLNet 0.6713 vs. TransUNet Hybrid 0.6585), cornea DSC varies by only 0.002, and iris DSC by 0.002. Because only a single training run was conducted per model, these differences are reported as single-seed point estimates rather than statistically tested architectural effects; however, the magnitude of the spread is small relative to the per-image DSC variability typically observed on small test sets of this kind.

On boundary precision (HD), ScLNet shows a small but consistent advantage on all three classes: cornea HD of 16.35 versus 21.01 (UNet) and 17.48 (TransUNet), and lesion HD of 7.30 versus 7.96 and 8.19 respectively. This advantage is consistent with ScLNet's boundary detection decoder, which explicitly optimizes for boundary accuracy through its auxiliary task.

The UNet baseline produced one complete-miss prediction (zero lesion pixels) on 1 of 77 test images, resulting in an undefined Hausdorff Distance that was excluded from averaging. The TransUNet Hybrid also produced one such case. ScLNet produced no complete misses, suggesting that the boundary decoder may provide modest detection reliability benefits.

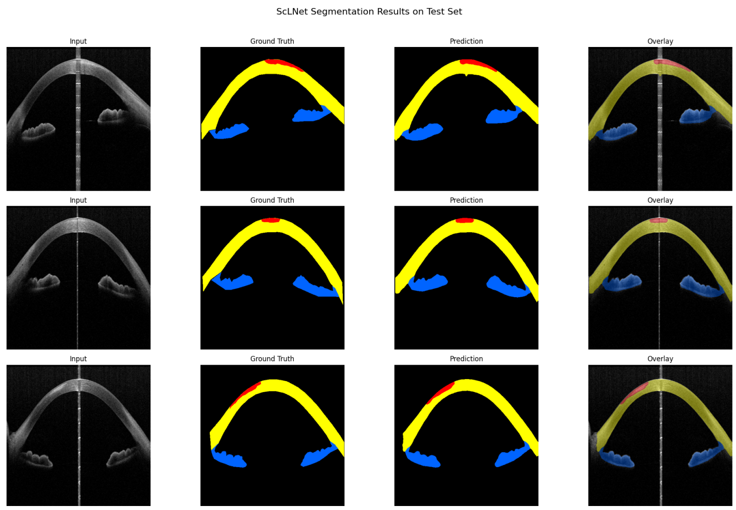

Figure 4: ScLNet qualitative results on representative test samples (3 rows shown). Columns: input, ground truth, prediction, overlay.

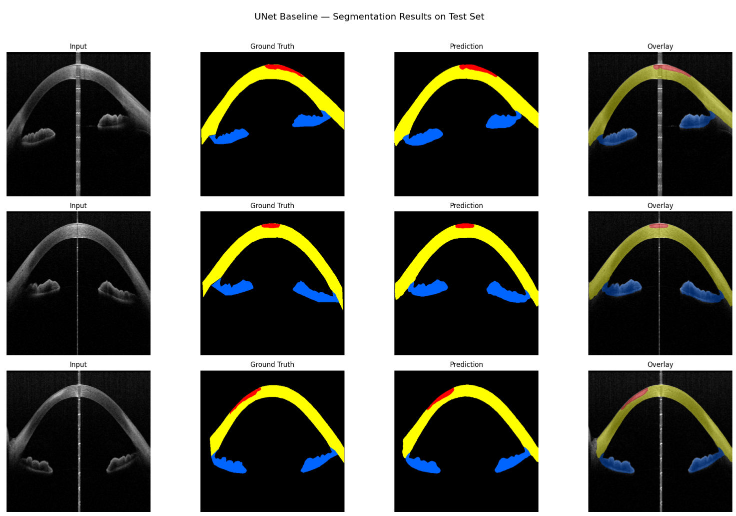

Figure 5: ResNet-34 UNet qualitative results on representative test samples (3 rows shown). Columns: input, ground truth, prediction, overlay.

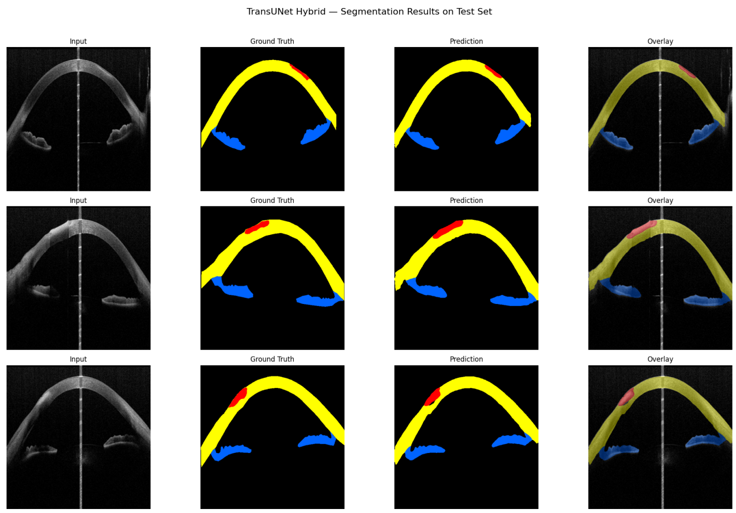

Figure 6: TransUNet Hybrid qualitative results on representative test samples (3 rows shown). Columns: input, ground truth, prediction, overlay.

5 Discussion

5.1 Architectural Contributions and Their Limits

The central observation of this study is that three architecturally distinct segmentation models converged to within 0.012 lesion DSC of each other on the AIDK keratitis dataset. This outcome is consistent with a data-limited ceiling on lesion segmentation, but it should be interpreted with care: all three models share a ResNet-34 encoder, were trained from scratch with the same SGD schedule, and were evaluated with a single training run per architecture. Convergence under these shared conditions does not rule out the possibility that larger architectural departures—for example, pretrained encoders, alternative backbone families, or different optimization regimes—could yield further gains. What it does suggest is that, within a reasonably broad family of encoder-decoder architectures trained under a standard protocol, lesion performance on AIDK is difficult to push past approximately 0.67 DSC.

ScLNet's innovations – multi-scale input pyramid, Dense Atrous Convolution blocks, and dual-task boundary decoder –provide measurable benefits only for boundary precision (Hausdorff Distance), not for region overlap (DSC). This is a nuanced finding: the boundary decoder and feature subtraction do capture boundary-relevant information, but this does not translate to improved region segmentation when the fundamental challenge is lesion ambiguity rather than boundary localization. The parameter overhead of ScLNet (92.6M versus 30M for UNet) is not justified by the performance difference on this dataset.

The TransUNet Hybrid, despite its global self-attention, boundary-aware features, asymmetric loss weighting, and deep supervision, achieved slightly *lower* lesion DSC (0.6585) than the simpler baselines. The result should not be viewed as evidence against the potential of architectural innovation to break the lesion ceiling. It reflects, rather, the limitations of the particular instantiation tested. First, the transformer bottleneck was trained from scratch on only 537 training images, whereas transformers typically require substantially more data or pretrained initialization to learn robust global patterns given their weaker inductive biases compared to CNNs [15]. Second, the Focal Tversky loss, while theoretically motivated for small-object segmentation, introduced training instability and degraded precision (0.6241 versus 0.6788 for ScLNet) without the intended gain in recall; the asymmetric false-negative penalty increased the model's tendency to predict lesion pixels, but these additional predictions were insufficiently accurate to improve DSC. Third, the model used the same SGD schedule as the CNN baselines, whereas transformer-based segmentation models commonly benefit from AdamW with warmup and different learning-rate schedules. A better-tuned transformer hybrid—with pretrained encoder weights, transformer-appropriate optimization, and a more stable loss—could plausibly yield different results. The failure of this particular model therefore speaks more to the difficulty of deploying transformers in small-data clinical regimes than to the architectural question itself; the architectural-convergence argument above rests primarily on the ScLNet-versus-UNet comparison rather than on this negative result.

5.2 The Lesion Segmentation Ceiling

The lesion DSC of approximately 0.67 should be interpreted in the context of established reference points for this dataset and for lesion segmentation more broadly:

- **AIDK human inter-annotator agreement:** The dataset authors report a mean Dice of 0.83 between expert annotators for lesion segmentation, with an intraclass correlation coefficient (ICC) of 0.708 [7]. The modest ICC indicates substantial disagreement between experts, implying that a significant portion of the gap between model performance (0.67) and human performance (0.83) may be attributable to annotation noise rather than true model deficiency.
- **General lesion segmentation IAA:** On the IMA++ dataset, the largest publicly available multi-annotator skin lesion segmentation dataset with 5,111 segmentations from 15 annotators, Abhishek et al. [20] report mean inter-annotator Dice scores of 0.791 ± 0.215 for benign lesions and 0.753 ± 0.227 for malignant lesions. The large standard deviations indicate that even for well-studied lesion types with extensive annotator pools, human consensus is highly variable. This suggests that annotation variability is an inherent challenge in lesion segmentation, not specific to the keratitis domain.

- **AIDK authors' unpublished result:** Sun et al. [7] report achieving a lesion DSC of 0.79 using an undisclosed "improved U-Net" architecture trained on the AIDK dataset supplemented with additional private data. This result was described as future work and has not been published. The supplementary private data is not publicly available, making this result non-reproducible and not directly comparable to these results, which use only the publicly released dataset.

These reference points suggest that the implemented models achieve approximately 81% of the human expert level (0.67/0.83) and approximately 85% of the authors' unpublished result (0.67/0.79), despite using only publicly available data and no supplementary training samples. Given the substantial inter-expert disagreement (ICC = 0.708), a meaningful portion of the remaining gap is likely irreducible without either improving the annotation process or fundamentally changing the evaluation methodology (e.g., evaluating against multi-annotator consensus rather than single expert labels).

5.3 Practical Findings for Future Work on AIDK

In addition to the primary architectural comparison, several practical insights emerged during this study that are relevant to future work on the AIDK dataset:

Loss function design. Per-class independent log-Dice loss proved critical for enabling lesion learning. The weighted-average formulation, where Dice scores across all classes are averaged before taking the log, suppresses gradient signal for minority classes. This finding is consistent with the broader literature on class-imbalanced segmentation [21] and represents a practical recommendation for anyone working with this dataset.

Iris label conflict. Combining the partial-frame and full-frame subsets introduces a systematic annotation inconsistency: iris tissue visible in partial-frame images is labeled as background because only cornea and lesion are annotated in that subset. Training with both subsets and iris as a segmentation class collapses iris DSC from 0.93 to 0.48. Future work using both subsets should either exclude iris as a segmentation target or implement label-aware masking that ignores iris predictions for partial-frame images during loss computation.

Class weighting is counterproductive with per-class Dice. Applying explicit class weights in addition to per-class Dice loss degrades performance for classes that are already well segmented and fails to benefit the target class. This behavior reflects an over-correction, as the per-class Dice objective inherently balances gradient contributions across classes independent of pixel prevalence.

5.4 Limitations

This study has several limitations. First, all experiments use only the publicly available AIDK dataset, which comprises images from a single OCT device (Tomey Casia 2). Performance on OCT images from other devices is unknown, and domain shift between devices is a recognized challenge in ophthalmic imaging [6]. Second, all models were trained from scratch without pretrained encoder weights, to maintain a controlled comparison; ImageNet pretraining could improve performance, particularly for the transformer model, but would introduce a confounding variable. Third, the test set contains only 77 images from 6 patients, limiting statistical power and potentially introducing patient-level biases. Fourth, the TransUNet Hybrid used the same SGD optimizer and learning rate schedule as the CNN-based models; transformer architectures may benefit from different optimization strategies (e.g., AdamW with warmup), which was not explored in order to maintain experimental control. Finally, only the full-frame subset was used for the three-way comparison; the recommended configuration of both subsets with iris excluded (maximizing lesion patient diversity to 58 patients) was not tested for all three architectures and remains an avenue for future work.

6 Conclusion

This work presents the first publicly reproducible deep learning benchmark for keratitis lesion segmentation on the AIDK AS-OCT dataset. Three architecturally distinct models were evaluated under controlled conditions: ScLNet, a multi-task boundary-aware architecture adapted from scleral lens OCT segmentation; a ResNet-34 UNet baseline that strips ScLNet's novel components; and a TransUNet Hybrid incorporating a transformer bottleneck, Focal Tversky loss, and deep supervision, designed specifically to address the lesion segmentation ceiling.

All three architectures converge to lesion DSC within 0.012 of each other (0.659–0.671), with cornea DSC exceeding 0.96 and iris DSC exceeding 0.92 across all models. The TransUNet Hybrid, despite its targeted architectural innovations, did not surpass the simpler baselines on lesion segmentation, although its particular instantiation—trained from scratch, optimized with SGD rather than transformer-typical optimization, and using a loss function that proved unstable in practice—was not a clean test of whether transformer-based approaches can break this ceiling. Taken together, the convergence of the two CNN-based models and the narrow overall spread across all three architectures are consistent with a lesion segmentation ceiling that is substantially data-limited on AIDK. At the same time, the shared ResNet-34 backbone, the absence of pretraining, and the use of a single training run per model leave this interpretation open to refinement with further experiments.

The achieved lesion DSC of approximately 0.67 represents 81% of the human expert inter-annotator agreement (0.83 DSC, ICC = 0.708) and 85% of the dataset authors' unpublished result with supplementary private data (0.79 DSC). Given the substantial variability among expert annotators, future work on AIDK should consider data-side interventions—such as expanding annotated patient cohorts, improving labeling consistency through multi-annotator consensus protocols, and adopting semi-supervised or self-supervised methods that exploit unlabeled clinical AS-OCT scans—alongside continued architectural and optimization exploration, particularly with pretrained encoders and transformer-appropriate training regimes that this study did not evaluate.

7 Code Availability

The source code for this study, including model implementations, training pipelines, and evaluation notebooks, is publicly available on GitHub: (Automated Segmetation of Corneal Structures and Keratitis Lesions in AS-OCT Images; accessed April 16, 2026.)

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